

## Analytics for Decision Making

### Power BI Analytical Dashboard

#### Power Query / ETL checklist

- Load Superstore CSV. Remove blank rows, trim text, fix types.
- Standardize text (=Text.Trim, proper case). Replace missing region/state with lookup.
- Create **DimDate**: expand from min(OrderDate)→max(ShipDate), create full date table with DateKey (Format yyyyMMdd). Mark as Date table later.
- Create **DimCustomer**: remove duplicates on CustomerID; add surrogate CustomerKey (Index).
- Create **DimProduct**: distinct ProductID/ProductName/SubCategory/Category; add ProductKey.
- Create **DimRegion**: distinct Region/State mapping.
- Create **FactSales**: keep transactional columns, replace natural keys by surrogates (merge to retrieve keys). Ensure numeric columns are typed as Decimal/Whole.
- Load to model. Set Date columns to Date type, mark DimDate as “Date table.”

DAX Measures (at least 5)

- **Total Sales**
- **Total Profit**
- **Average Order Value**
- **Profit Margin**
- **YoY or MoM growth**
- **Sales per customer**

#### Relationships (cardinality & direction)

- FactSales[OrderDateKey] → DimDate[DateKey] (many-to-one, single direction)
- FactSales[ShipDateKey] → DimDate[DateKey] (role-playing, create separate relation or duplicate DimDate as ShipDate dimension or use inactive relation + USERELATIONSHIP)
- FactSales[CustomerKey] → DimCustomer[CustomerKey] (many-to-one)

## Analytics for Decision Making

- FactSales[ProductKey] → DimProduct[ProductKey] (many-to-one)
  - FactSales[RegionKey] → DimRegion[RegionKey] (many-to-one)
- Set cross-filter direction to Single unless you explicitly need bi-directional.

## Hierarchies

- Date hierarchy: Year > Quarter > Month > Day
- Product: Category > SubCategory > ProductName
- Geography: Region > State > City

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name |
|--------|----------------|------------|------------|----------------|-------------|---------------|
| 1      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | Second Class   | CG-12520    | Claire e      |
| 2      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | Second Class   | CG-12520    | Claire e      |
| 3      | CA-2016-138688 | 6/12/2016  | 6/16/2016  | Second Class   | DV-13045    | Darrin        |
| 4      | US-2015-108966 | 10/11/2015 | 10/16/2015 | Standard Class | SO-20335    | Sean C        |
| 5      | US-2015-108966 | 10/11/2015 | 10/16/2015 | Standard Class | SO-20335    | Sean C        |
| 6      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosius       |
| 7      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosius       |
| 8      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosius       |
| 9      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosius       |
| 10     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosius       |
| 11     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosius       |
| 12     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosius       |
| 13     | CA-2017-114412 | 4/15/2017  | 4/20/2017  | Standard Class | AA-10480    | Andrew        |
| 14     | CA-2016-161389 | 12/5/2016  | 12/10/2016 | Standard Class | IM-15070    | Irene h       |
| 15     | US-2015-118983 | 11/22/2015 | 11/26/2015 | Standard Class | HP-14815    | Harold        |
| 16     | US-2015-118983 | 11/22/2015 | 11/26/2015 | Standard Class | HP-14815    | Harold        |
| 17     | CA-2014-105893 | 11/11/2014 | 11/16/2014 | Standard Class | PK-19075    | Pete Ki       |
| 18     | CA-2014-167164 | 5/13/2014  | 5/15/2014  | Second Class   | AG-10270    | Alejan        |
| 19     | CA-2014-143336 | 8/27/2014  | 9/1/2014   | Second Class   | ZD-21925    | Zuschu        |
| 20     | CA-2014-143336 | 8/27/2014  | 9/1/2014   | Second Class   | ZD-21925    | Zuschu        |
| 21     | CA-2014-143336 | 8/27/2014  | 9/1/2014   | Second Class   | ZD-21925    | Zuschu        |
| 22     | CA-2016-137330 | 12/9/2016  | 12/13/2016 | Standard Class | KB-16585    | Ken Bl        |
| 23     | CA-2016-137330 | 12/9/2016  | 12/13/2016 | Standard Class | KB-16585    | Ken Bl        |
| 24     | US-2017-156909 | 7/16/2017  | 7/18/2017  | Second Class   | SF-20065    | Sandra        |
| 25     | CA-2015-106320 | 9/25/2015  | 9/30/2015  | Standard Class | EB-13870    | Emily e       |

## 3. Analysis by Category and Subcategory

- The subcategories **Supplies**, **Bookcases**, and **Tables** do not generate profit. They are likely to lead to debt.
- Conversely, the subcategories **Copiers**, **Phones**, and **Accessories** are the most profitable.
- Overall, **the Technology category is the best represented. Therefore, this is the one that should be promoted.** However, **the Furniture category requires review.**

# Analytics for Decision Making

## Detailed Profitability (Drill-through)

This page allows a user to right-click any product and get an instant , detailed report on its profitability across different region with the Total Sales/ Total Profit and Profit Margin. The green data increased being top sales than red.

Drill-through — Detailed Profitability

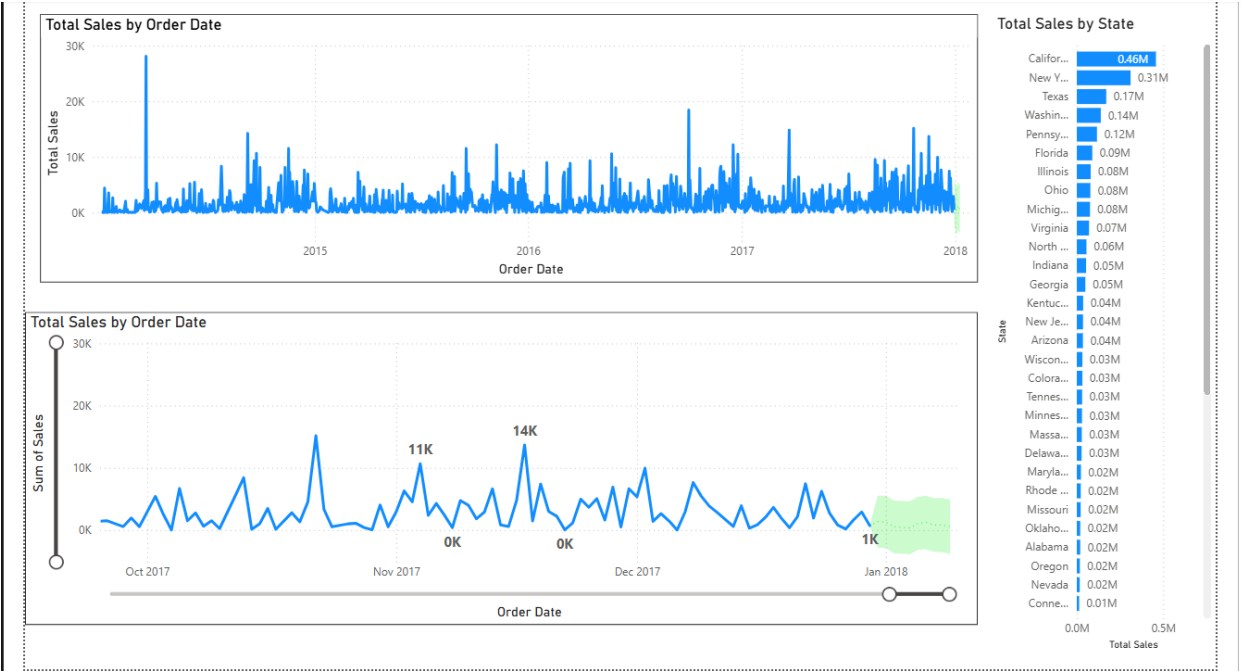
## Sales & Profit Overview

- High-level operational snapshot.
- Visuals: Cards for total sales/profit, slicers, trend charts.

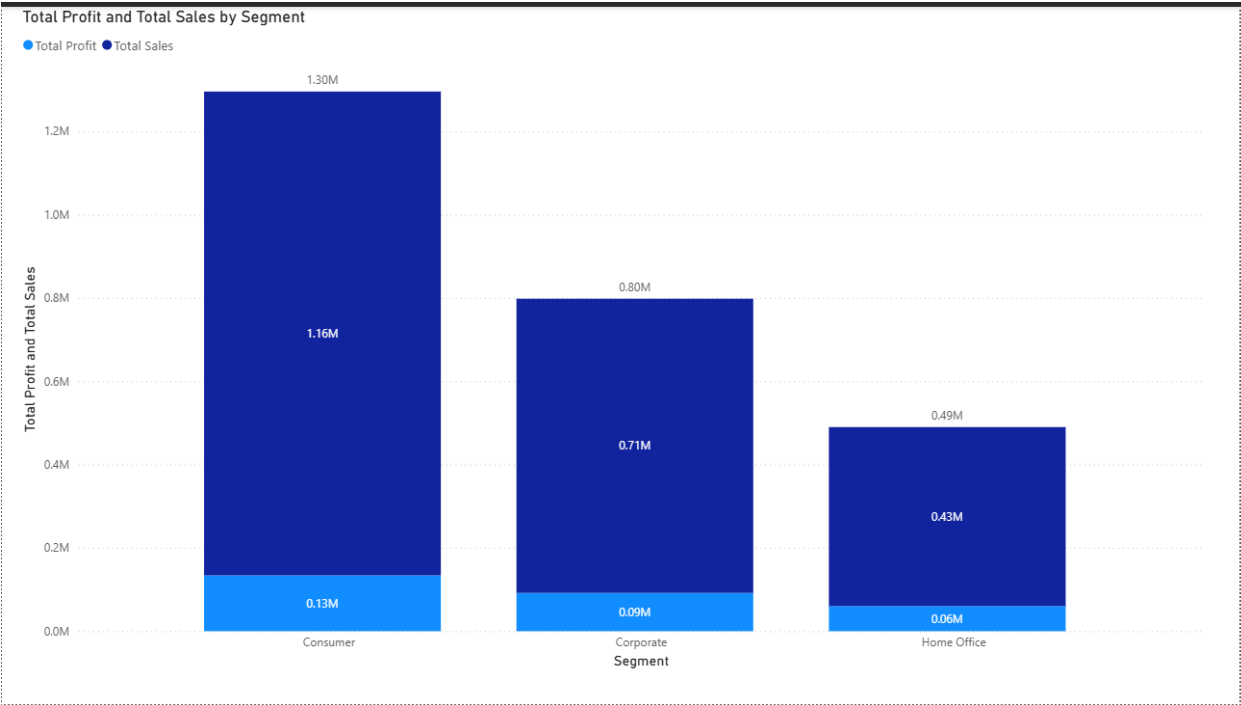
| Order ID       | Year | Quarter | Month     | Day | Customer Name     | Product Name                                                                                    | Sum of Quantity | Sum of Sales | Sum |
|----------------|------|---------|-----------|-----|-------------------|-------------------------------------------------------------------------------------------------|-----------------|--------------|-----|
| CA-2014-100006 | 2014 | Qtr 3   | September | 7   | Dennis Kane       | AT&T EL51110 DECT                                                                               | 3               | 377.97       |     |
| CA-2014-100090 | 2014 | Qtr 3   | July      | 8   | Ed Braxton        | Hon 2111 Invitation Series Corner Table                                                         | 3               | 502.49       |     |
| CA-2014-100090 | 2014 | Qtr 3   | July      | 8   | Ed Braxton        | Wilson Jones Ledger-Size, Piano-Hinge Binder, 2", Blue                                          | 6               | 196.70       |     |
| CA-2014-100293 | 2014 | Qtr 1   | March     | 14  | Neil Französisch  | Xerox 1887                                                                                      | 6               | 91.06        |     |
| CA-2014-100328 | 2014 | Qtr 1   | January   | 28  | Jasper Cacioppo   | Pressboard Covers with Storage Hooks, 9 1/2" x 11", Light Blue                                  | 1               | 3.93         |     |
| CA-2014-100363 | 2014 | Qtr 2   | April     | 8   | Jim Mitchum       | Binder Clips by OIC                                                                             | 2               | 2.37         |     |
| CA-2014-100363 | 2014 | Qtr 2   | April     | 8   | Jim Mitchum       | Things To Do Today Spiral Book                                                                  | 3               | 19.01        |     |
| CA-2014-100391 | 2014 | Qtr 2   | May       | 25  | Barry Weirich     | Strathmore Photo Frame Cards                                                                    | 2               | 14.62        |     |
| CA-2014-100678 | 2014 | Qtr 2   | April     | 18  | Kunst Miller      | Cameo Buff Policy Envelopes                                                                     | 3               | 149.35       |     |
| CA-2014-100678 | 2014 | Qtr 2   | April     | 18  | Kunst Miller      | DMI Arturo Collection Mission-style Design Wood Chair                                           | 3               | 317.06       |     |
| CA-2014-100678 | 2014 | Qtr 2   | April     | 18  | Kunst Miller      | Kensington Expert Mouse Optical USB Trackball for PC or Mac                                     | 3               | 227.98       |     |
| CA-2014-100678 | 2014 | Qtr 2   | April     | 18  | Kunst Miller      | Prang Dustless Chalk Sticks                                                                     | 2               | 2.69         |     |
| CA-2014-100706 | 2014 | Qtr 4   | December  | 16  | Laurel Elliston   | Case Logic 2.4GHz Wireless Keyboard                                                             | 2               | 99.98        |     |
| CA-2014-100706 | 2014 | Qtr 4   | December  | 16  | Laurel Elliston   | Ultra Door Push Plate                                                                           | 6               | 29.46        |     |
| CA-2014-100762 | 2014 | Qtr 4   | November  | 24  | Nat Gilpin        | Adams Telephone Message Book w/Frequently-Called Numbers Space, 400 Messages per Book           | 2               | 15.96        |     |
| CA-2014-100762 | 2014 | Qtr 4   | November  | 24  | Nat Gilpin        | Dot Matrix Printer Tape Reel Labels, White, 5000/Box                                            | 2               | 196.62       |     |
| CA-2014-100762 | 2014 | Qtr 4   | November  | 24  | Nat Gilpin        | Hunt PowerHouse Electric Pencil Sharpener, Blue                                                 | 4               | 151.92       |     |
| CA-2014-100762 | 2014 | Qtr 4   | November  | 24  | Nat Gilpin        | Xerox 1885                                                                                      | 3               | 144.12       |     |
| CA-2014-100860 | 2014 | Qtr 1   | March     | 26  | Cindy Stewart     | Smead Alpha-Z Color-Coded Name Labels First Letter Starter Set                                  | 5               | 18.75        |     |
| CA-2014-100867 | 2014 | Qtr 4   | October   | 19  | Eugene Hildebrand | RCA Visys Integrated PBX 8-Line Router                                                          | 6               | 321.55       |     |
| CA-2014-100881 | 2014 | Qtr 1   | March     | 28  | Daniel Raglin     | AT&T TR1909W                                                                                    | 3               | 302.38       |     |
| CA-2014-100895 | 2014 | Qtr 2   | June      | 2   | Stewart Visinsky  | Hot File 7-Pocket, Floor Stand                                                                  | 2               | 356.94       |     |
| CA-2014-100895 | 2014 | Qtr 2   | June      | 2   | Stewart Visinsky  | Mophie Juice Pack Helium for iPhone                                                             | 3               | 239.97       |     |
| CA-2014-100895 | 2014 | Qtr 2   | June      | 2   | Stewart Visinsky  | Sanford Colorific Scented Colored Pencils, 12/Pack                                              | 2               | 8.56         |     |
| CA-2014-100916 | 2014 | Qtr 4   | October   | 21  | Frank Hawley      | Eldon Base for stackable storage shelf, platinum                                                | 5               | 194.70       |     |
| CA-2014-100916 | 2014 | Qtr 4   | October   | 21  | Frank Hawley      | Hon 2111 Invitation Series Straight Table                                                       | 4               | 591.32       |     |
| CA-2014-100916 | 2014 | Qtr 4   | October   | 21  | Frank Hawley      | SANFORD Liquid Accent Tank-Style Highlighters                                                   | 1               | 2.84         |     |
| CA-2014-100972 | 2014 | Qtr 4   | November  | 19  | Dennis Bolton     | White Dual Perf Computer Printout Paper, 2700 Sheets, 1 Part, Heavyweight, 20 lbs., 14 7/8 x 11 | 3               | 166.44       |     |
| CA-2014-101147 | 2014 | Qtr 4   | December  | 2   | Matt Collins      | Staple holder                                                                                   | 1               | 2.39         |     |
| CA-2014-101175 | 2014 | Qtr 4   | December  | 9   | Dario Medina      | Acco Perma 3000 Stacking Storage Drawers                                                        | 6               | 100.70       |     |
| Total          |      |         |           |     |                   |                                                                                                 | 37873           | 2,297,200.86 | 286 |

Analytics for Decision Making

Forecast of Total Sales by date. The California state showing is the highest the sales with 0.42M



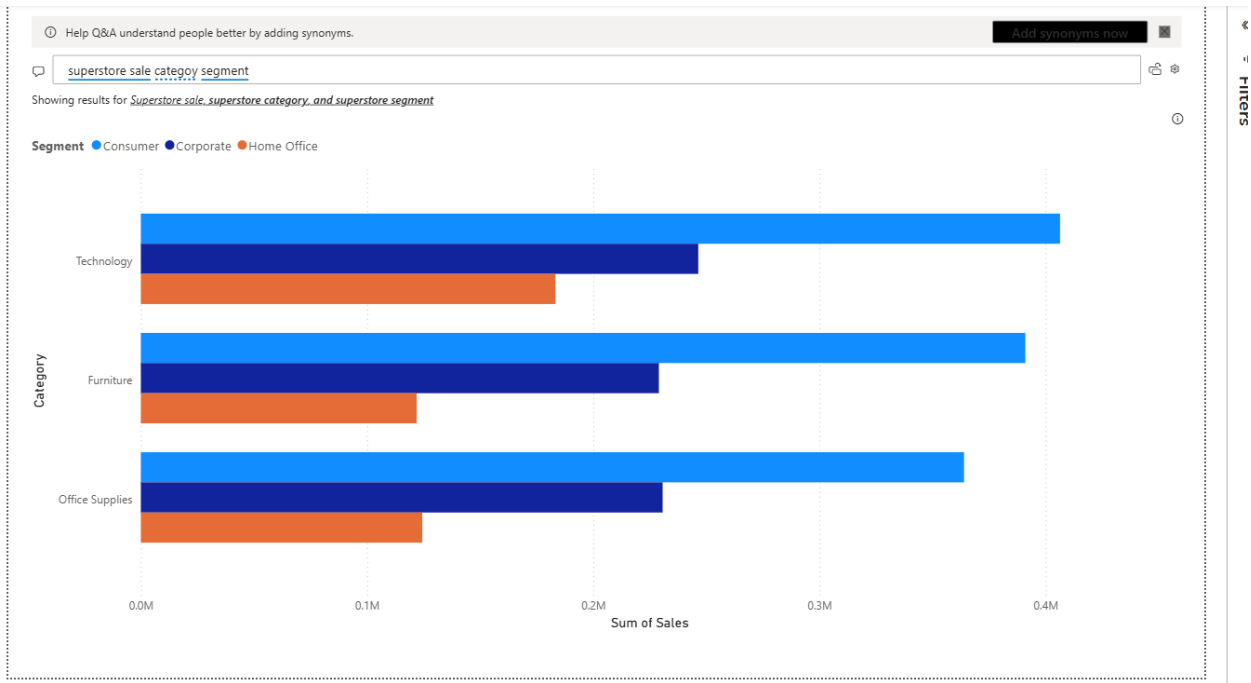
Total Profit and Total Sales by Segment



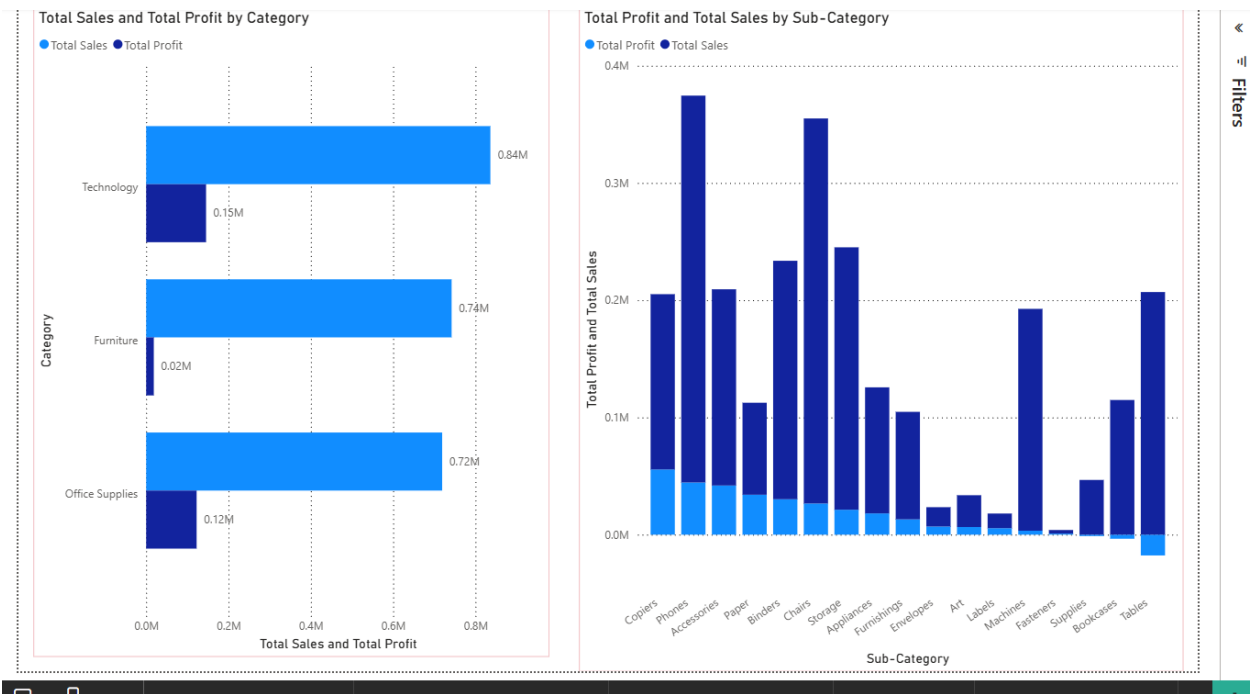
# Analytics for Decision Making

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## Q&A question : To type in dashboard



## Total Sales and Total Profit by Category



# Analytics for Decision Making

## Decomposition Tree

